

Mohammed Thaha Dawood

Senior Backend & AI Engineer (Node.js | AWS | Machine Learning) — Trivandrum, India
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Summary

Senior Backend Engineer with 2+ years of experience building scalable cloud-native SaaS platforms and AI-powered systems. Strong expertise in Node.js, AWS, microservices architecture, and REST API development, with hands-on experience in Machine Learning and Data Science. Proven ability to design data pipelines, train and deploy ML models, and optimize system performance, cost efficiency, and production reliability.

Technical Skills

Node.js, TypeScript, NestJS, Express.js, Python, Django, FastAPI, REST APIs, WebSockets, AWS ECS, AWS Lambda, AWS RDS, AWS IAM, AWS CloudWatch, Docker, Terraform, Jenkins, GitHub Actions, Infrastructure as Code, CI/CD Pipelines, Microservices Architecture, Event-Driven Systems

MongoDB, PostgreSQL, Redis, Kafka, Database Design, Indexing, Sharding, Query Optimization, Caching Strategies

Machine Learning, Data Science, Pandas, NumPy, Scikit-learn, XGBoost, Feature Engineering, Model Training, Model Evaluation (MAE, RMSE, R²), Time-Series Forecasting, Fraud Detection, Data Pipelines, ETL

Natural Language Processing, Sentiment Analysis, FinBERT, RoBERTa, Streamlit, Dash

Experience

Team Lead / Backend Engineer Feb 2024 – Present
Cygnonex Innovation Pvt Ltd, Kochi

- Led backend architecture for a multi-tenant ERP SaaS platform serving 500+ businesses with 99.9% uptime.
- Designed and deployed 9+ Node.js microservices using Docker, AWS ECS, and REST APIs.
- Improved API latency by 40% through Redis caching, query optimization, and indexing strategies.
- Reduced AWS infrastructure costs by 25% via container optimization and autoscaling.
- Integrated ML-driven analytics and reporting features into core ERP workflows.
- Built CI/CD pipelines using Jenkins and GitHub Actions for automated testing and deployment.

Python Developer Intern Sep 2023 – Feb 2024
Lexyle Technologies, Trivandrum

- Developed Django REST APIs processing over 1M+ records per week.
- Built data processing and ETL scripts to support analytics and reporting.
- Improved database performance by 75% using Redis caching and optimized indexing.

AI / Machine Learning Projects

Aegis — Real-Time Fraud Detection System

GitHub: github.com/MohdThaha/Aegis-Fraud-Detection

Built an end-to-end ML pipeline for fraud detection with synthetic data generation, feature engineering, and model training using Scikit-learn and XGBoost. Deployed real-time inference APIs using FastAPI and Docker, with model insights visualized via Streamlit.

DataLens — Sales Forecasting and Demand Prediction

GitHub: github.com/MohdThaha/DataLens

Developed regression-based ML models for sales forecasting using historical transaction data. Implemented feature engineering and model evaluation (MAE, RMSE, R²) to generate actionable business insights.

Pulse — Real-Time Financial Sentiment Analysis Platform

GitHub: github.com/MohdThaha/Pulse-Real-time-Market-Sentiment-Analyzer

Built a real-time NLP system to analyze financial market sentiment from Twitter/X and Reddit data. Implemented text preprocessing, sentiment analysis using FinBERT/RoBERTa, and exposed inference via FastAPI with live dashboards in Streamlit/Dash.

Education

B.Tech in Computer Science Engineering 2019 – 2023
Valia Koonambaikulathamma College of Engineering and Technology